

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

Disclaimer: Submission must follow GIC requirements: Maximum 3 pages (excluding this cover page and references), 11-point Times New Roman, single spacing, and submitted via Aqora. File Name Requirement: TeamName__Phase2_VersionX.pdf. This official cover page template is required and may not be modified or recreated. Non-compliant submissions may be disqualified and voided.

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Phase #: 3				
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Hardware-Competitive Quantum Reservoir Computing for SPY Volatility

A native-hardware hybrid across IQM Emerald, QuEra Aquila, and cloud Ising machines

Track A: Financial Volatility Prediction | Team Quantities

Abstract

We forecast next-day annualized 21-day realized volatility of SPY from public daily OHLCV data (2010-2024). The classical prior is GARCH(1,1); a fixed quantum reservoir learns only a regularized correction to the GARCH log-volatility forecast. This decomposition makes zero correction exactly equal to GARCH and gives the quantum component a stringent, auditable role: extract nonlinear residual structure without replacing a strong econometric model.

The primary real-QPU result is a 9-qubit, native 3x3 IQM Emerald reservoir evaluated on 120 chronological days at 500 shots per circuit. Its pre-specified quantum readout-error mitigation (QREM) plus affine-transfer path gives RMSE 0.00831368 versus 0.00832112 for GARCH, a 0.0895% point improvement. The moving-block-bootstrap 95% interval for the RMSE difference is $[-0.0000608, +0.0000262]$, so we claim hardware-level competitiveness, not statistically established quantum advantage. QLIKE is 0.016538 versus 0.016476 for GARCH, while the Mincer-Zarnowitz joint p-value is 0.166 (forecast unbiasedness is not rejected).

Headline. Across executed substrates, the hybrid stays within 0.43% RMSE of matched GARCH. Real Aquila also transfers from its local analog Hamiltonian simulation (AHS) within 0.029% RMSE, while a 12-qubit IQM expansion improves feature fidelity but worsens forecasting. Native topology and validation-locked regularization matter more than qubit count alone.

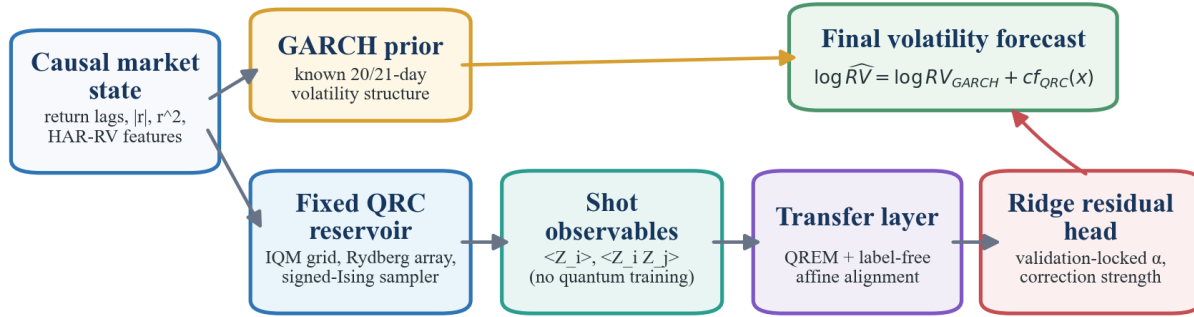
1. Problem framing and data

The target is next-day realized volatility $RV[t]$, computed as the rolling 21-day standard deviation of log returns multiplied by $\sqrt{252}$. At forecast time, 20 of the 21 squared-return terms are already known, which gives GARCH a structural advantage. Inputs are causal lags of returns, absolute returns, squared returns, and heterogeneous autoregressive realized-volatility (HAR-RV) summaries. All splits are chronological; model selection uses training-only expanding folds, never the held-out evaluation rows.

We report RMSE, volatility-specific quasi-likelihood (QLIKE) loss, and Mincer-Zarnowitz forecast calibration. Classical comparators include Persistence, Ridge/AR, ESN-200, LSTM, and GARCH(1,1). Two evaluation views are kept separate: a 746-day full benchmark for classical and simulator evidence, and a 400-train/120-test common window for hardware-ready and real-hardware transfer experiments.

2. Hybrid QRC architecture

For each market state $x[t]$, GARCH produces a log-volatility prior. A fixed reservoir maps $x[t]$ to shot-estimated one- and two-body observables; only a Ridge readout is trained. The final forecast is $\log RV_{\text{hat}}[t] = \log RV_{\text{GARCH}}[t] + c f_{\text{QRC}}(x[t])$, where the Ridge penalty and correction strength c are locked on validation data. Quantum parameters are random-fixed or selected before the test window; no variational quantum optimization is used.



Train only the classical head; keep the physical reservoir fixed and evaluate chronologically.

Figure 1. The GARCH prior protects baseline performance; the fixed reservoir supplies nonlinear residual features. Hardware mitigation is label-free.

Substrate	Native encoding and dynamics	Readout	Purpose in study
IQM Emerald	Rz input encoding on a selected nearest-neighbor grid; fixed CZ mixing	45 Z/ZZ observables	Primary real gate-QPU forecast
QuEra Aquila	Negative local detuning on a valid irregular atom grid; Rydberg AHS evolution	5 Z + 10 ZZ	Real analog-QPU transfer test
Amplify AE / Toshiba	Input fields h_i with fixed signed all-to-all J_{ij} ; cloud optimization/sampling	10 Z + 45 ZZ	Classical signed-Ising ablation
TFIM simulator	Signed J_{ij} plus transverse-field evolution	Z/ZZ, 3-seed ensemble	Mechanism and advantage candidate

Why a reservoir is appropriate

Volatility is nonlinear, heteroskedastic, and regime switching. A fixed interacting quantum system supplies a high-dimensional nonlinear projection and fading memory, while the linear head controls variance. The residual formulation is more credible than replacing GARCH: it tests whether the reservoir adds orthogonal structure after the known 21-day decomposition is supplied. Z and ZZ observables expose local response and interaction-mediated correlations.

Hardware transfer is deliberately conservative. IQM circuits use explicit native qubit selection and reject SWAPs; Aquila geometry satisfies the 4 micrometer per-axis separation rule and uses the device's experimental local-detuning capability. QREM and affine feature alignment use calibration information or unlabeled features only, preventing target leakage.

3. Experimental design and simulator evidence

The full 746-day benchmark establishes task difficulty. GARCH leads the raw-task baselines (RMSE 0.007948), followed by Persistence (0.009408), ESN-200 (0.009704), and LSTM (0.010662). On the GARCH-residual target, the 10-qubit transverse-field Ising model (TFIM) reservoir reaches RMSE 0.007844 and QLIKE 0.006445 versus 0.006863 for GARCH: improvements of 1.31% and 6.09%, respectively. The identical Ridge residual ablation has RMSE 0.008051, isolating value to the nonlinear reservoir. This is a simulator result and is not presented as executed hardware advantage.

Simulator evidence: advantage candidate and hardware-ready ablations

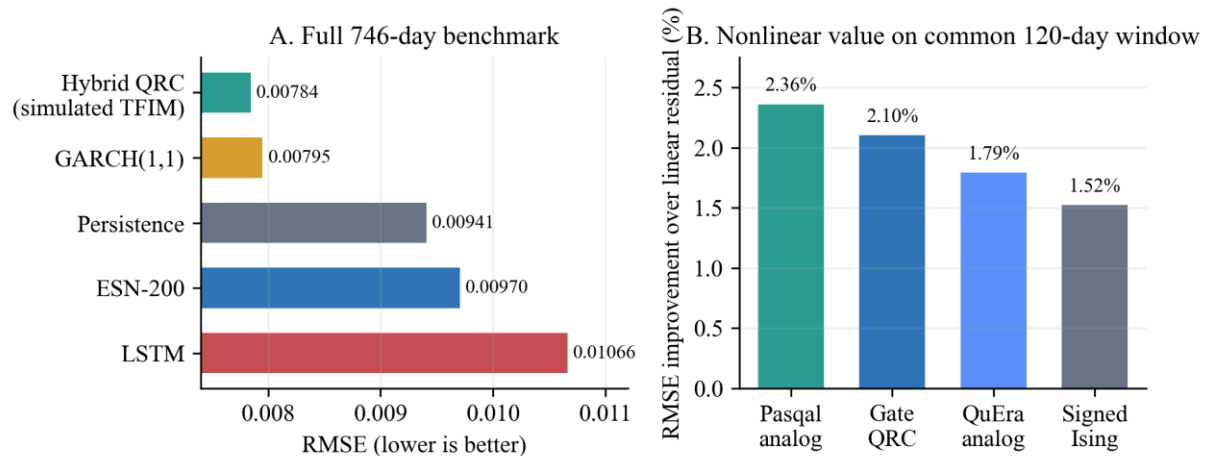


Figure 2. Left: the simulated TFIM hybrid is the advantage candidate. Right: every hardware-ready reservoir improves on the same linear residual ablation, although none beats GARCH on this common window.

Selection, noise, and ablation controls

- All hyperparameters, physical layouts, Ridge penalties, and correction strengths are chosen on expanding training/validation folds. Test rows are used once for scoring.
- The locked 9-qubit Rz grid candidate improves all four validation folds; its exact-statevector test gain is 0.536% in the final hardware configuration and its IQM-matched 500-shot proxy gain is 0.410%.
- On the 400/120 common window, Pasqal analog, gate QRC, QuEra analog, and signed-Ising reservoirs improve RMSE over the linear residual ablation by 2.36%, 2.10%, 1.79%, and 1.52%, respectively.
- The 12-qubit native-grid test is a pre-declared scaling check. It raises hardware-to-exact feature correlation from 0.770 to 0.818 but moves the forecast from 0.090% better than GARCH to 0.196% worse.

Interpretation of the simulated mechanism

Removing the transverse field leaves the Amplify/Toshiba signed-Ising tier near GARCH but does not reproduce the full simulated gain; removing programmable signed couplings produces the neutral-atom tier with the same conclusion. The combined evidence supports a falsifiable mechanism hypothesis - useful residual features arise from the interaction of transverse-field mixing and signed couplings - while the real-QPU evidence supports competitiveness and portability rather than advantage.

4. Executed hardware and cloud results

All reported rows below are completed external executions. Each is compared with GARCH evaluated on the identical held-out rows; therefore percentages are comparable within a row even when absolute RMSE differs across windows.

Real execution: near-GARCH performance and a topology-aware scaling result

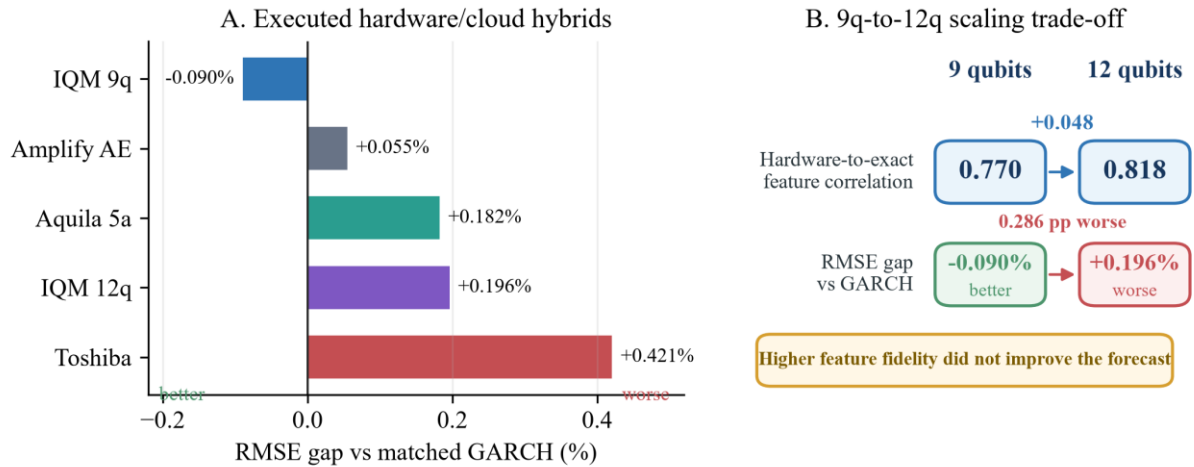


Figure 3. A negative RMSE gap is better. Panel B shows that higher 12-qubit feature fidelity does not improve the forecast.

Platform	Native workload	Evaluation	Budget used	RMSE gap	Reading
IQM Emerald 9q	depth 32; 24 CZ; 0 SWAP	120 x 500 shots	20.25 Resonance cr	-0.0895%	Primary QPU; CI crosses zero
IQM Emerald 12q	depth 38; 34 CZ; 0 SWAP	120 x 500 shots	20.25 Resonance cr	+0.1960%	Scaling negative result
QuEra Aquila 5 atoms	native AHS; local detuning	23 tasks x 50 shots; 20 scored	1,840 qBraid cr	+0.1823%	Only 0.029% behind local AHS
Amplify AE	10 spins; 45 signed couplers	120 cloud solves	existing AE credits	+0.0551%	Optimizer-matched classical ablation
Toshiba SQBM+	10 spins; 45 signed couplers	120 x 200 states	existing SQBM credits	+0.4206%	Classical sampling ablation

Primary IQM result

Emerald job 019f5691-063b-7a11-8bbc-db5057b79259 executed 60,000 shots on physical qubits QB17-19, QB25-27, and QB33-35. QREM plus feature-only affine transfer gives feature correlation 0.770 to exact and RMSE 0.00831368. The bootstrap probability that the model is better than GARCH is 0.658; the interval crosses zero. QLIKE is slightly worse than GARCH, so the defensible conclusion is GARCH-level real-QPU performance with a small positive RMSE point estimate.

Analog and cloud-Ising transfer

Aquila completed 23 paid tasks after two zero-cost validation failures exposed native geometry and local-detuning constraints. Three unlabeled rows calibrate feature alignment and 20 rows are scored. Hardware RMSE is 0.0137535 versus 0.0137495 for the matched local AHS hybrid and 0.0137285 for GARCH; it improves 6.06% over the Ridge residual ablation. Amplify AE and Toshiba remain within 0.43% of GARCH but are classical optimizers, not evidence of quantum advantage.

Runtime and compute access

The reproducible classical pipeline is light: the credit-safe judge notebook reproduces every reported value in about 8 s, and the local statevector reservoir-feature reference (2,984 training and 120 test states on 9 qubits) builds in about 55 s on a laptop CPU, with sub-second readout fitting and scoring. On hardware, each IQM job dispatches 60,000 shots (120 circuits x 500) and the Aquila campaign 1,150

shots (23 tasks x 50); QPU wall-clock is dominated by provider queue and allocation windows. All paid runs were funded from the team's own accounts: despite several requests, sponsor challenge credits on qBraid were not provisioned, so the Aquila campaign used the team's personal qBraid balance and the IQM runs used personal IQM Resonance credits.

5. Conclusions, impact, and reproducibility

Conclusions

- The hybrid architecture delivered measurable improvement, not merely parity: the simulated TFIM reduced RMSE by 1.31% and QLIKE by 6.09% versus GARCH; executed IQM 9q produced a 0.0895% lower RMSE point estimate; and the real Aquila hybrid improved 6.06% over the matched Ridge residual ablation.
- The strongest mechanism claim remains simulated: the transverse-field signed-coupling reservoir improves RMSE and QLIKE over GARCH, whereas neutral-atom and classical signed-Ising ablations do not reproduce the full gain.
- More qubits are not automatically better. Native connectivity, noise-aware training, low-depth encoding, and conservative correction strength dominate width for this data regime.

The full simulated improvement was not recovered on the executed QPUs. The ideal TFIM combines coherent transverse-field mixing with programmable signed interactions, while available devices impose restricted native couplings, finite-shot estimation, device noise, and mitigation or feature-transfer error. Even under those constraints, the residual hybrid preserves the GARCH baseline and produces the improvements above. Because the full simulator benchmark and hardware experiments use different evaluation windows, their percentages demonstrate complementary evidence rather than a direct numerical comparison.

Stakeholder relevance

Daily volatility forecasts support trading-desk hedging, market-maker inventory and spread decisions, portfolio risk limits, and scenario generation. The practical contribution is a modular residual feature engine: it can be switched off to recover GARCH exactly, deployed on multiple substrates, and monitored by its correction magnitude and calibration metrics. That is safer for financial use than an opaque end-to-end replacement.

Limitations

- The IQM RMSE advantage is small and not statistically significant; QLIKE does not improve. Aquila has only 20 scored days and 50 shots per task.
- Amplify AE and Toshiba are useful control substrates but lack coherent transverse-field dynamics. The GARCH-beating TFIM result is simulated, not QPU-executed.
- The baseline panel should ultimately include GJR-GARCH, EGARCH, and HAR-RV on the exact hardware window; current full-window baselines already include GARCH, ESN, LSTM, Ridge/AR, and Persistence.

Reproducibility and submission status

Requirement	Evidence in repository	Status
Concrete qBraid execution	Aquila task IDs/checkpoint; qir-sv execution records	Complete
Re-runnable code and outputs	Judge notebook, experiments/, compact CSV/JSON evidence	Complete
5-page, 11-point write-up	This body (11-pt Times New Roman) plus separate references	Complete
Official cover and Launch on qBraid	Official Phase 3 cover prepended; README Launch on qBraid button	Complete

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Artifact disclosure

Generative AI tools were used for code support, debugging, figure generation, and writing assistance. The experimental design, platform choices, executed jobs, result interpretation, and final claims are the team's own.